
Technical Documentation of the Human Resources for Health Gap Projection Model

Methodology, Model Structure, Assumptions, Outputs and Limitations

Application: HRH Gap Tool
Platform: R Shiny
Projection horizon: User-defined, default 2035
Target year: User-defined, default 2030
Cadres modelled: Doctors, nurses/midwives, pharmacists and dentists

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1 Overview

The Human Resources for Health (HRH) Gap Tool is an interactive analytical application developed in R Shiny to estimate present and future health workforce shortages under alternative planning scenarios. The tool projects health workforce supply, estimates workforce requirements using benchmark-based and health-need-linked methods, and quantifies the resulting workforce gaps and coverage levels.

The tool is intended to support strategic workforce planning, policy dialogue, scenario analysis and priority setting. It generates projections for the following health workforce cadres:

- Doctors;
- Nurses and midwives;
- Pharmacists; and
- Dentists.

Results may be generated for one or more countries and presented through target-year summaries, trend plots, workforce rankings and downloadable files.

2 Model Objectives

The model is designed to:

1. estimate historical workforce growth rates;
2. project future workforce supply under alternative scenarios;
3. estimate workforce requirements using benchmark and need-linked approaches;
4. quantify workforce shortages, surpluses and coverage levels;
5. identify the year in which projected workforce supply meets estimated requirements; and
6. support comparison of countries, cadres and policy scenarios.

3 Overall Model Framework

The model consists of four main analytical components:

1. data validation and preprocessing;
2. workforce supply projection;
3. workforce requirement estimation; and
4. workforce gap and coverage analysis.

The conceptual framework of the model is illustrated in Figure 1.

4 Input Data Requirements

As part of the input data requirements, the uploaded dataset must contain the variables listed in Table 1.

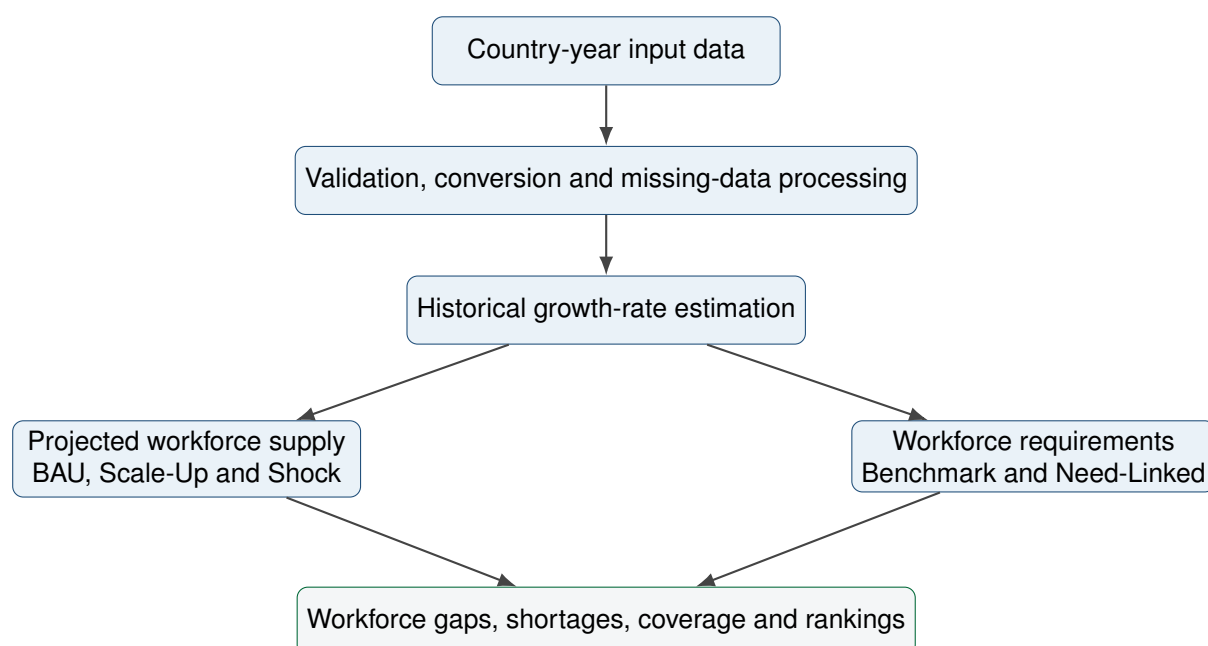


Figure 1: Conceptual structure of the HRH gap model.

Table 1: Required input variables

Variable	Description
ISO	Country identifier or ISO code.
Country	Country name.
Year	Observation year.
Population	Total national population.
GDP	Gross domestic product or another consistently defined economic measure.
Doctors	Doctors per 10,000 population.
Nursing_Midwifery	Nurses and midwives per 10,000 population.
Dentists	Dentists per 10,000 population.
Pharmacists	Pharmacists per 10,000 population.
Doctors_Number	Total number of doctors.
Nursing_Midwifery_Number	Total number of nurses and midwives.
Dentists_Number	Total number of dentists.
Pharmacists_Number	Total number of pharmacists.
DALYS	Disability-adjusted life years or a consistently defined DALY measure.
MMR	Maternal mortality ratio.
NCDMR	Non-communicable disease mortality rate.
UHC_SCI	Universal Health Coverage Service Coverage Index.

All variables used in calculations are converted to numeric format where applicable. Country identifiers are treated as character variables, and year is treated as an integer.

5 Data Processing

5.1 Workforce Stock Estimation

Where an observed workforce headcount is available, the model uses the reported headcount directly. Where the headcount is missing, it is estimated from workforce density and population:

$$S_{c,t,j} = D_{c,t,j} \times \frac{P_{c,t}}{10,000}, \quad (1)$$

where:

- $S_{c,t,j}$ is the workforce stock for country c , year t and cadre j ;
- $D_{c,t,j}$ is the workforce density per 10,000 population; and
- $P_{c,t}$ is the population.

5.2 Missing Workforce Data

Complete workforce time series are required to estimate historical workforce growth rates and project future workforce supply. However, workforce datasets may contain missing observations because of incomplete reporting, changes in information systems, unavailable administrative records, or differences in reporting frequency.

To reduce the loss of country-cadre observations during model estimation, missing workforce stock values are handled separately for each country and cadre using a sequential three-stage procedure:

1. linear interpolation for missing values between two observed years;
2. forward filling for missing values occurring after an observed value; and
3. backward filling for missing values occurring before the first observed value.

This procedure is implemented in the model using the following R code:

```
group_by(ISO, Country, cadre) %>%
mutate(
  stock = zoo::na.approx(
    stock,
    x = Year,
    na.rm = FALSE
  )
) %>%
mutate(
  stock = zoo::na.locf(
    stock,
    na.rm = FALSE
  )
) %>%
mutate(
  stock = zoo::na.locf(
    stock,
    fromLast = TRUE,
```

```

    na.rm = FALSE
  )
)

```

The three methods are applied sequentially. Each stage therefore acts only on values that remain missing after the preceding stage.

5.2.1 Linear Interpolation

The model first applies linear interpolation to missing workforce observations that occur between two known values. This method assumes that workforce stock changes at a constant rate between the two observed years.

Suppose workforce stock is observed in years t_1 and t_2 , but is missing in an intermediate year t . The missing value is estimated as:

$$S_t = S_{t_1} + \frac{t - t_1}{t_2 - t_1} (S_{t_2} - S_{t_1}), \quad (2)$$

where:

- S_t is the interpolated workforce stock in year t ;
- S_{t_1} is the observed workforce stock in the earlier year;
- S_{t_2} is the observed workforce stock in the later year; and
- $t_1 < t < t_2$.

For example, suppose the number of doctors is 100 in 2018 and 120 in 2020, while the value for 2019 is missing. The interpolated value is:

$$S_{2019} = 100 + \frac{2019 - 2018}{2020 - 2018} (120 - 100) = 110. \quad (3)$$

Thus, the completed series becomes:

Table 2: Example of linear interpolation

Year	Workforce stock
2018	100
2019	110
2020	120

Linear interpolation is implemented using:

```
zoo::na.approx(stock, x = Year, na.rm = FALSE)
```

The option `na.rm = FALSE` ensures that missing observations at the beginning or end of a time series are retained for processing in the subsequent steps.

5.2.2 Forward Filling

After interpolation, the model applies forward filling to remaining missing values. Forward filling, also known as the last observation carried forward method, replaces a missing value with the most recent preceding observed value.

For a missing value in year t , the method can be represented as:

$$S_t = S_{t-1}, \quad (4)$$

where S_{t-1} is the most recent non-missing workforce stock.

For example:

Table 3: Example of forward filling

Year	Original value	Filled value
2020	130	130
2021	Missing	130
2022	Missing	130

This operation is implemented using:

```
zoo::na.locf(stock, na.rm = FALSE)
```

The approach assumes that workforce stock remains unchanged during the period for which no new observation is available.

In the current model, forward filling is applied to the historical workforce series before growth rates are estimated. It should not be confused with the future supply projection, which is subsequently generated using the estimated workforce growth rate.

5.2.3 Backward Filling

The final step addresses missing values occurring before the first observed workforce value. Backward filling replaces these missing values with the earliest available subsequent observation.

For a missing value before the first observed year:

$$S_t = S_{t+1}, \quad (5)$$

where S_{t+1} represents the earliest subsequent non-missing value.

For example:

Table 4: Example of backward filling

Year	Original value	Filled value
2016	Missing	95
2017	Missing	95
2018	95	95

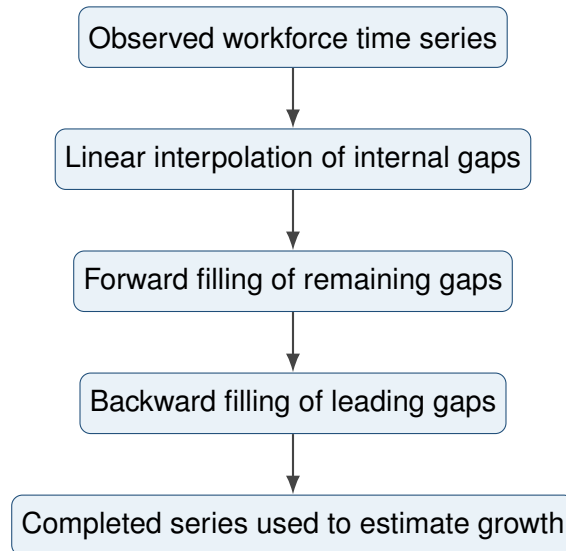
This operation is implemented using:

```
zoo::na.locf(
  stock,
  fromLast = TRUE,
  na.rm = FALSE
)
```


The argument `fromLast = TRUE` reverses the usual last-observation-carried-forward procedure and carries the earliest available observation backward.

5.2.4 Order of Application

The imputation methods are applied in the following order:



This order is important because linear interpolation is preferred for internal gaps, where information is available on both sides of the missing observation. Forward and backward filling are used only for observations that cannot be interpolated.

5.2.5 Role in the Workforce Projection Model

The completed workforce series is used to estimate historical annual workforce growth rates. For each country and cadre, the model calculates:

$$g_t = \frac{S_{t+1} - S_t}{S_t}. \quad (6)$$

The most recent valid growth rates are then averaged to obtain the Business-as-Usual growth rate. Consequently, imputed workforce values may influence:

- the annual historical growth rates;
- the estimated Business-as-Usual growth rate;
- the Scale-Up and Shock scenario growth rates;
- projected workforce supply;
- projected workforce gaps; and
- estimated workforce coverage.

The missing-data procedure is therefore not merely for displaying complete data. It contributes directly to the workforce supply projections.

5.2.6 Underlying Assumptions

The missing-data procedure relies on the following assumptions:

1. Workforce stock changes approximately linearly between two observed years.
2. Missing internal observations can be reasonably estimated from neighbouring values.
3. The most recent observed workforce stock is an acceptable approximation for subsequent missing historical years.
4. The earliest observed workforce stock is an acceptable approximation for preceding missing years.
5. Missingness does not result from major structural changes that would make interpolation or value carrying inappropriate.
6. The definitions and measurement of workforce cadres remain consistent across years.

5.2.7 Limitations

Although this procedure creates complete time series, it has several limitations.

- Linear interpolation may conceal genuine annual fluctuations.
- Forward filling assumes no workforce change during missing periods.
- Backward filling assumes that earlier workforce levels were equal to the first observed value.
- The procedure may not capture abrupt changes arising from recruitment campaigns, migration, industrial action, policy reforms, conflict, epidemics or reporting-system changes.
- Long sequences of missing observations may produce highly uncertain imputed values.
- Imputed observations are treated in the growth-rate calculation in the same way as observed values.

For these reasons, the proportion and pattern of imputed values should be assessed before interpreting the projections. Sensitivity analyses may also be conducted by excluding countries or cadres with substantial missing data or by comparing results under alternative imputation methods.

5.2.8 Recommended Reporting

For transparency, the final analysis should report:

- the number and percentage of workforce observations that were missing;
- the number filled through interpolation;
- the number filled through forward filling;
- the number filled through backward filling;
- the maximum length of consecutive missing periods; and
- any countries or cadres with extensive imputation.

A sensitivity analysis comparing projections based on the imputed series with projections restricted to observed values would strengthen the assessment of model robustness. **In the current analysis, this was not reported.**

5.3 Missing Health-Need Indicators

The need-linked workforce requirement model relies on four health-need indicators: Disability-Adjusted Life Years (DALYs), Maternal Mortality Ratio (MMR), Non-Communicable Disease Mortality Rate (NCDMR), and the Universal Health Coverage Service Coverage Index (UHC SCI). Since these indicators are not always available for every country and year, missing values are imputed to enable the estimation of a complete health-need index throughout the projection period.

Within each country, missing observations are first replaced using the most recent available value (Last Observation Carried Forward, LOCF). Where missing values occur before the first recorded observation, the earliest available value is carried backward (Next Observation Carried Backward, NOCB). This approach preserves the most recent evidence on a country's health profile while avoiding unnecessary exclusion of countries or years from the analysis.

Unlike workforce stock, these health indicators are not projected using trend models. Instead, the most recent observed values are assumed to remain constant during the projection period unless updated data become available. This assumption reflects the fact that reliable long-term forecasts for these indicators are generally unavailable within the application and ensures that projected changes in workforce requirements arise primarily from demographic growth and the user-defined need-adjustment parameters rather than uncertain projections of disease burden.

The imputed health-need indicators are subsequently standardized and combined to construct the composite health-need index used in the need-linked workforce requirement model. Consequently, the imputation procedure directly influences the estimated health-need index and the resulting workforce requirements under the need-based approach.

The underlying assumption is that short-term changes in DALYs, MMR, NCD mortality, and UHC service coverage are relatively gradual and that the most recent available observations provide reasonable approximations in the absence of updated data. One should therefore interpret need-linked projections cautiously for countries with substantial missing data and, where possible, update the input dataset with the latest available health indicators.

6 Workforce Supply Projection Model

6.1 Historical Annual Growth

Historical workforce growth is estimated separately for each country and health workforce cadre using the available workforce stock time series. The annual growth rate is calculated as the proportional change in workforce stock between two consecutive years:

$$g_{c,t,j} = \frac{S_{c,t+1,j} - S_{c,t,j}}{S_{c,t,j}}, \quad (7)$$

where $S_{c,t,j}$ denotes the workforce stock for country c , year t , and cadre j . Positive values indicate workforce expansion, while negative values indicate a decline in workforce stock.

To reduce the influence of short-term fluctuations, the model estimates the Business-as-Usual (BAU) growth rate as the arithmetic mean of the most recent valid annual growth observations. Annual growth is calculated only when the current workforce stock is positive and the subsequent stock is available. Non-finite growth rates and rates equal to or below -100% are excluded.

If m valid annual growth rates are available, where $m \leq k$, the BAU growth rate is:

$$\bar{g}_{c,j}^{BAU} = \frac{1}{m} \sum_{i=1}^m g_{c,i,j}. \quad (8)$$

The parameter k is user-defined, with a default value of five, and represents the maximum number of recent valid annual growth observations used rather than necessarily five calendar years. If fewer than k valid rates are available, all valid rates are used. Where no valid growth rate can be estimated, a growth rate of zero is assumed.

6.2 Supply Projection Equation

Future workforce supply is projected using compound annual growth:

$$\hat{S}_{c,t,j,s} = S_{c,t_0,j} (1 + g_{c,j,s})^{t-t_0}, \quad (9)$$

where:

- t_0 is the last observed year;
- $S_{c,t_0,j}$ is the workforce stock in the last observed year;
- $g_{c,j,s}$ is the annual growth rate for scenario s ; and
- $t - t_0$ is the number of projection years.

If no valid historical workforce growth rate is available, the model assigns a growth rate of zero for projection purposes.

6.3 Projection Scenarios

To evaluate the potential impact of different workforce planning strategies, the model projects workforce supply under three alternative scenarios: Business as Usual (BAU), Scale-Up, and Shock. Each scenario modifies the annual workforce growth rate while keeping the underlying projection model unchanged. These scenarios are intended to support policy analysis by illustrating how changes in workforce growth may influence future workforce availability, shortages, and coverage.

6.3.1 Business as Usual (BAU)

The Business-as-Usual scenario assumes that future workforce growth continues at the same average rate observed during the recent historical period. The annual growth rate is therefore defined as:

$$g_{c,j,BAU} = \bar{g}_{c,j}^{BAU}. \quad (10)$$

This scenario serves as the baseline against which alternative policy interventions are compared. It assumes that existing patterns of workforce recruitment, training, attrition, and retention continue without major changes.

6.3.2 Scale-Up

The Scale-Up scenario represents an accelerated workforce expansion resulting from enhanced investments or policy interventions, such as increased training capacity, recruitment, retention strate-

gies, or improved financing for human resources for health. The annual growth rate is adjusted upward as:

$$g_{c,j,ScaleUp} = \bar{g}_{c,j}^{BAU} + \delta_{up}, \quad (11)$$

where δ_{up} is a user-defined positive increment. The default value is 0.01, corresponding to a one-percentage-point increase in the annual workforce growth rate. Larger values of δ_{up} represent more ambitious workforce expansion strategies.

6.3.3 Shock

The Shock scenario simulates adverse conditions that slow workforce growth, such as economic downturns, budget constraints, health emergencies, migration of health professionals, or increased workforce attrition. The annual growth rate is therefore reduced according to:

$$g_{c,j,Shock} = \bar{g}_{c,j}^{BAU} + \delta_{shock}, \quad (12)$$

where δ_{shock} is a user-defined negative adjustment. The default value is -0.01 , corresponding to a one-percentage-point reduction in annual workforce growth. More negative values represent increasingly severe disruptions to workforce expansion.

The three scenarios are deterministic and differ only in the assumed annual growth rate. They do not explicitly model the mechanisms driving workforce changes, but instead provide a flexible framework for exploring the potential consequences of alternative workforce growth assumptions on projected supply, workforce gaps, and coverage over the projection period.

6.3.4 Growth-Rate Validation and Projection Safeguards

For each country, cadre and scenario, the application verifies that the projected annual growth rate satisfies:

$$g_{c,j,s} > -1. \quad (13)$$

A scenario producing annual growth equal to or below -100% is rejected because it would generate invalid workforce projections. Projected workforce stock is additionally constrained to remain non-negative:

$$\hat{S}_{c,t,j,s} = \max \left[0, S_{c,t_0,j} (1 + g_{c,j,s})^{t-t_0} \right]. \quad (14)$$

No upper cap is imposed on positive growth rates. Users should therefore review implausibly high historical rates and scenario adjustments before interpreting the results.

6.4 Projected Workforce Density

Although workforce supply is projected as the total number of health workers, comparisons across countries require a standardized measure that accounts for differences in population size. The model therefore converts projected workforce stock into workforce density, expressed as the number of health workers per 10,000 population.

Thus, projected workforce density is calculated as:

$$\hat{D}_{c,t,j,s} = \frac{\hat{S}_{c,t,j,s}}{P_{c,t}} \times 10,000, \quad (15)$$

where:

- $\hat{D}_{c,t,j,s}$ is the projected workforce density for country c , year t , cadre j , and projection scenario s ;
- $\hat{S}_{c,t,j,s}$ is the projected workforce stock;
- $P_{c,t}$ is the projected population for year t ; and
- the multiplier 10,000 expresses workforce availability per 10,000 population.

Workforce density provides a standardized measure of workforce availability that enables meaningful comparisons across countries and over time, irrespective of population size. It is widely used in health workforce planning because it relates workforce capacity to the population potentially requiring health services.

Within the HRH Gap Tool, projected workforce density is calculated internally as part of the supply-projection process. It supports comparison of workforce stock relative to population and can be compared with benchmark densities. In the current interface, however, the principal displayed and downloadable outputs are projected headcounts, requirements, gaps, additional staff needed and coverage measures. Projected density is not currently provided as a separate downloadable table.

An increase in workforce density over time may result from growth in workforce supply, slower population growth, or both, whereas a decline may reflect insufficient workforce expansion relative to population growth. Consequently, workforce density should be interpreted alongside projected workforce stock and workforce requirements to provide a comprehensive assessment of future workforce adequacy.

7 Population and GDP Projections

7.1 Compound Annual Growth Rate (CAGR)

Population and GDP are projected separately for each country using the first and last valid observations within the recent historical window defined by:

$$t \geq t_{\max} - k. \quad (16)$$

Depending on data availability, this window may contain up to $k + 1$ annual observations. CAGR is calculated only when at least two valid positive observations are available and the final year is later than the initial year.

For a generic variable X , the CAGR is:

$$r_X = \left(\frac{X_{last}}{X_{first}} \right)^{\frac{1}{T}} - 1, \quad (17)$$

where T is the number of years between the first and last observations.

Future values are projected as:

$$\hat{X}_t = X_{last}(1 + r_X)^{t-t_{last}}. \quad (18)$$

If the estimated CAGR is unavailable, the model assumes zero growth.

Although GDP is projected in the application, **it is not currently included directly in the workforce requirement formula**. It is retained as a potential contextual or future model input/expansion.

8 Country-Specific Model Specifications

The application can apply common default specifications to all selected countries or allow the user to edit country-specific values. The following parameters may therefore vary by country:

- need-adjustment strength, λ_c ;
- cadre-specific benchmark densities, $B_{c,j}$;
- need-linked cadre shares, $q_{c,j}$;
- Scale-Up adjustment, $\delta_{c,up}$; and
- Shock adjustment, $\delta_{c,shock}$.

When common defaults are selected, the same initial values are assigned to every selected country. When country-specific editing is enabled, the values in the country specification table are used independently for each country.

8.1 Specification Validation

Before the model is executed, the application verifies that:

1. each selected country appears once in the specification table;
2. no country is duplicated;
3. λ_c is non-negative;
4. all benchmark densities are non-negative;
5. the four cadre shares sum to one for each country;
6. the scenario adjustments are numeric; and
7. all selected countries have complete specification records.

The cadre-share condition is:

$$q_{c,Doctors} + q_{c,NursesMidwives} + q_{c,Pharmacists} + q_{c,Dentists} = 1. \quad (19)$$

The model is not executed when these validation requirements are not met.

9 Workforce Requirement Models

The application estimates workforce requirements using two alternative approaches:

1. benchmark-based requirements; and
2. need-linked requirements.

9.1 Benchmark-Based Requirement

For each cadre, the user specifies a benchmark density $B_{c,j}$ per 10,000 population. The required headcount is:

$$R_{c,t,j}^{Benchmark} = B_{c,j} \times \frac{P_{c,t}}{10,000}. \quad (20)$$

The default benchmark densities are shown in Table 5.

Table 5: Default benchmark densities per 10,000 population

Cadre	Default benchmark
Doctors	1.5
Nurses and midwives	20.0
Pharmacists	2.0
Dentists	0.5

These values are user-configurable and should be justified using national norms, international benchmarks, service-delivery requirements or policy targets.

9.2 Need-Linked Requirement

The need-linked model adjusts the total benchmark workforce density according to a composite health-need index.

9.2.1 Standardization of Need Indicators

Each indicator is standardized within country:

$$Z_{c,t,m} = \frac{X_{c,t,m} - \bar{X}_{c,m}}{s_{c,m}}, \quad (21)$$

where:

- $X_{c,t,m}$ is indicator m for country c in year t ;
- $\bar{X}_{c,m}$ is the within-country mean; and
- $s_{c,m}$ is the within-country standard deviation.

If the standard deviation is zero or unavailable, standardized values are set to zero.

Where an indicator has no variation across the completed country time series, all its standardized values are therefore zero. Such an indicator contributes nothing to variation in the composite need index, even when its absolute level is high.

Health indicators are carried forward into the projection years before the within-country mean and standard deviation are calculated. Consequently, the selected projection horizon can influence the standardized values because repeated future observations contribute to the standardization parameters.

9.2.2 Composite Health-Need Index

The model combines several dimensions of population health need into a single composite health-need index. The index is constructed as a weighted sum of standardized values for Disability-Adjusted Life Years (DALYs), Maternal Mortality Ratio (MMR), Non-Communicable Disease Mortality Rate (NCDMR), and Universal Health Coverage Service Coverage Index (UHC SCI):

$$N_{c,t} = w_D Z_{DALYS,c,t} + w_M Z_{MMR,c,t} + w_N Z_{NCDMR,c,t} - w_U Z_{UHC,c,t}, \quad (22)$$

where $N_{c,t}$ denotes the composite health-need index for country c in year t , and Z represents the standardized value of each indicator. Standardization places indicators measured on different scales onto a common scale, allowing them to be combined meaningfully.

The indicator weights satisfy:

$$w_D + w_M + w_N + w_U = 1, \quad (23)$$

with each weight constrained to lie between zero and one. The weights determine the relative contribution of each indicator to the overall index and may be adjusted by the user to reflect different policy priorities or analytical assumptions.

DALYs, maternal mortality, and NCD mortality enter the index positively because higher values indicate a greater burden of disease and, therefore, greater potential demand for health services and health workers. UHC service coverage enters negatively because higher coverage is interpreted as evidence of stronger service availability and lower unmet health-system need.

The composite index provides a measure of relative health need over time within each country. Because each country is modelled separately and each indicator is standardized using that country's own completed time series, the resulting index is not a directly comparable cross-country health-need score. A higher value indicates that health need in a given year is greater than that country's own time-series average, while a lower value indicates need below its own average.

Within the model, the index acts as a scaling factor in the need-linked workforce requirement approach. Years with higher health-need index values within a country generate larger need-linked workforce adjustments for that country. Cross-country comparisons of need-linked requirements should therefore be interpreted cautiously because the standardized index does not use a single pooled reference distribution.

The index should be interpreted cautiously because its value depends on the selected indicators, their standardization, the assigned weights, and the completeness of the underlying data. Sensitivity analysis using alternative weighting schemes is therefore recommended to assess how strongly the projected workforce requirements depend on these assumptions.

9.2.3 Need-Adjusted Total Density

The total benchmark density is:

$$B_{total,c} = \sum_j B_{c,j}. \quad (24)$$

The need-adjusted total workforce density is:

$$D_{c,t}^{Need} = \max[0, B_{total,c} + \lambda_c N_{c,t}], \quad (25)$$

where λ_c is the country-specific parameter controlling the strength of the need adjustment.

The non-negativity constraint prevents the estimated workforce requirement from falling below zero.

9.2.4 Need-Linked Total Requirement

The total need-linked workforce requirement is:

$$R_{c,t}^{Need,total} = D_{c,t}^{Need} \times \frac{P_{c,t}}{10,000}. \quad (26)$$

9.2.5 Allocation Across Cadres

The total need-linked requirement is allocated to each cadre using user-defined shares:

$$R_{c,t,j}^{Need} = q_{c,j} R_{c,t}^{Need,total}, \quad (27)$$

subject to:

$$\sum_j q_{c,j} = 1. \quad (28)$$

The default cadre shares are shown in Table 6.

Table 6: Default cadre shares in the need-linked model

Cadre	Default share
Doctors	0.20
Nurses and midwives	0.70
Pharmacists	0.07
Dentists	0.03

10 Workforce Gap and Coverage Measures

10.1 Workforce Gap

For each country, year, cadre, requirement model and scenario:

$$G_{c,t,j,r,s} = \hat{S}_{c,t,j,s} - R_{c,t,j,r}, \quad (29)$$

where r identifies the benchmark or need-linked requirement model.

Interpretation:

- $G < 0$: projected workforce shortage;
- $G = 0$: projected supply equals requirement; and
- $G > 0$: projected workforce surplus.

10.2 Additional Staff Needed

The number of additional workers required is:

$$A_{c,t,j,r,s} = \max \left(0, R_{c,t,j,r} - \hat{S}_{c,t,j,s} \right). \quad (30)$$

This measure is restricted to non-negative values. A surplus therefore produces an additional requirement of zero.

10.3 Coverage Ratio

Coverage is calculated as:

$$C_{c,t,j,r,s} = \frac{\hat{S}_{c,t,j,s}}{R_{c,t,j,r}}. \quad (31)$$

Coverage percentage is:

$$C_{c,t,j,r,s}^{\%} = 100 \times C_{c,t,j,r,s}. \quad (32)$$

Interpretation:

- $C < 1$: projected supply is below the requirement;
- $C = 1$: projected supply meets the requirement; and
- $C > 1$: projected supply exceeds the requirement.

Coverage is not calculated where the requirement is zero or unavailable.

11 Threshold-Year Identification

For selected coverage and supply-versus-requirement displays, the application identifies the first projected year in which coverage reaches or exceeds one:

$$C_{c,t,j,r,s} \geq 1. \quad (33)$$

The threshold year is defined as:

$$t^* = \min \{t : C_{c,t,j,r,s} \geq 1\}. \quad (34)$$

The threshold is identified from annual projection points and is not interpolated between years. If coverage does not reach one during the selected projection horizon, no threshold year is reported.

12 Model Parameters

Table 7: Main model parameters

Parameter	Default	Interpretation
Historical growth window (k)	5 years	Maximum number of recent valid annual workforce growth observations used to estimate BAU growth.
Projection horizon	2035	Final year of the workforce projection.
Target year	2030	Year used for target summaries and rankings.
Need adjustment strength (λ_c)	2.5	Controls the change in required workers per 10,000 population associated with a one-unit change in the composite health-need index.
Scale-Up increment	0.01	Percentage-point increase added to BAU annual growth.
Shock adjustment	-0.01	Percentage-point reduction added to BAU annual growth.
Need indicator weights	0.25 each	Relative contribution of DALYs, MMR, NCD mortality and UHC coverage to the need index.
Cadre shares	0.20, 0.70, 0.07, 0.03	Allocation of total need-linked requirements across cadres.

12.1 Projection-Year Validation

The application requires that:

$$t_{\text{horizon}} \geq t_{\text{latest observed}}, \quad (35)$$

$$t_{\text{target}} \leq t_{\text{horizon}}, \quad (36)$$

and:

$$t_{\text{target}} \geq t_{\text{latest observed}}. \quad (37)$$

These checks ensure that the target-year summary falls within the projection period and is not defined before the latest observed year in the loaded dataset. When countries have different latest observed years, the dashboard reports the range of country-specific base years.

13 Model Outputs

The application produces the following outputs:

- target-year workforce supply and requirements;

- additional staff needed by country, cadre, model and scenario;
- workforce coverage trends;
- workforce gap trends;
- supply-versus-requirement plots;
- approximate years in which supply meets requirements;
- country rankings by shortages, gaps or coverage;
- downloadable target-year tables in CSV format;
- downloadable full gap datasets in CSV format;
- downloadable country specification tables in CSV format; and
- downloadable copies of the loaded dataset in RDS format.

13.1 Country Ranking Classification

The ranking display classifies country results according to the selected metric:

- additional staff needed greater than zero is classified as a shortage;
- additional staff needed equal to zero is classified as surplus or balance;
- a negative workforce gap is classified as a shortage;
- a non-negative workforce gap is classified as surplus or balance;
- coverage below 100% is classified as a shortage; and
- coverage equal to or above 100% is classified as surplus or balance.

This metric-specific logic prevents positive coverage values or positive workforce gaps from being incorrectly labelled as shortages.

14 Model Assumptions

14.1 Data Assumptions

1. Reported workforce headcounts and densities are sufficiently accurate for projection.
2. Workforce density measures refer to comparable cadres and definitions across countries and years.
3. Where headcounts are missing, workforce density and population can be used to reconstruct approximate headcounts.
4. Internal missing workforce observations can be reasonably estimated through linear interpolation.
5. Forward and backward filling of remaining missing observations does not introduce material bias.
6. Health-need indicators are consistently defined over time.
7. Population and GDP data are measured consistently across the projection base period.

14.2 Supply Projection Assumptions

1. Recent historical workforce growth is informative about future growth under BAU.
2. The arithmetic mean of recent annual growth rates adequately represents the underlying workforce growth process.
3. Annual workforce growth remains constant within each scenario over the projection period.
4. Workforce growth compounds annually.
5. Scale-Up and Shock conditions can be represented by fixed percentage-point changes to BAU growth.
6. Where historical growth cannot be estimated, zero growth is a reasonable fallback assumption.
7. Historical growth rates equal to or below -100% are excluded.
8. Scenario-specific annual growth rates equal to or below -100% are rejected.
9. Projected workforce stock is constrained to remain non-negative.
10. The model does not impose an upper limit on positive workforce growth.

14.3 Population and Economic Assumptions

1. Population and GDP follow constant compound annual growth rates estimated from recent observations.
2. The effect of demographic structure, age distribution, urbanization and epidemiological transition is not modelled separately.
3. GDP is not currently used directly in the requirement calculation.

14.4 Benchmark Requirement Assumptions

1. User-defined benchmark densities represent appropriate workforce planning targets.
2. Benchmark densities remain constant throughout the projection period.
3. Workforce requirements increase proportionally with population.
4. The benchmark model does not directly account for differences in disease burden, service utilization, geography or productivity.

14.5 Need-Linked Requirement Assumptions

1. DALYs, maternal mortality, NCD mortality and UHC service coverage are appropriate indicators of health-system need.
2. Higher DALYs, MMR and NCD mortality imply greater workforce need.
3. Higher UHC service coverage implies lower additional workforce need.
4. Standardized within-country indicator values provide a meaningful measure of relative need over time.
5. The selected weights correctly represent the relative importance of the four need indicators.
6. The need index is additive and linear.

7. The lambda parameter appropriately translates standardized health need into changes in workforce density.
8. The need-linked total workforce requirement can be divided across cadres using fixed shares.
9. Cadre shares remain constant across countries and years unless altered by the user.

14.6 Workforce System Assumptions

The model does not explicitly simulate:

- workforce training pipelines;
- graduation and recruitment delays;
- migration;
- retirement;
- mortality;
- resignation and other forms of attrition;
- unemployment or underemployment;
- absenteeism;
- dual practice;
- public-private sector movement;
- subnational distribution;
- rural-urban imbalances;
- differences in working hours;
- workforce productivity;
- changing scopes of practice;
- task shifting;
- technological change; or
- changes in models of service delivery.

These factors are assumed to be reflected only indirectly in historical workforce growth.

15 Limitations

The HRH Gap Projection Tool has some limitations that should be considered when interpreting the results.

1. **Trend extrapolation:** Workforce, population, and GDP projections are based on recent historical trends and assume constant annual growth within each scenario. Unexpected changes in policy, financing, workforce production, migration, or labour-market conditions are not explicitly modelled.
2. **Data quality and completeness:** The accuracy of projections depends on the quality of the underlying data. Missing, inconsistent, or misclassified workforce and health indicator data

may affect estimated growth rates, workforce requirements, and projected gaps, even after interpolation and imputation.

3. **Simplified workforce requirements:** Benchmark-based requirements rely on fixed staffing-density targets, while the need-linked model assumes a fixed cadre skill mix and a composite health-need index. These assumptions may not fully capture country-specific service delivery needs or changes in workforce composition over time.
4. **Relative health-need index:** The composite health-need index is standardized separately within each country and therefore represents relative changes over time rather than an absolute or directly comparable measure of health need across countries.
5. **Deterministic projections:** The model produces deterministic scenario-based projections and does not quantify uncertainty through confidence intervals, prediction intervals, or probabilistic sensitivity analyses.

16 Interpretation of Results

The outputs should be interpreted as scenario-based planning estimates rather than precise forecasts. Differences between BAU, Scale-Up and Shock scenarios illustrate the possible effect of alternative workforce growth pathways.

A projected shortage indicates that workforce supply is below the selected benchmark or need-linked requirement. It does not necessarily indicate the exact number of workers who should immediately be recruited because workforce planning must also consider:

- fiscal space;
- training capacity;
- absorptive capacity;
- deployment and retention;
- productivity;
- service-delivery organization; and
- subnational distribution.

Policy conclusions should therefore be supported by sensitivity analysis and consultation with workforce planners, professional bodies, ministries of health, training institutions and labour-market experts.

17 Recommended Sensitivity Analyses

To assess robustness of this model, one should vary:

- the historical growth window;
- benchmark densities;
- Scale-Up and Shock adjustments;
- need indicator weights;
- the lambda parameter;

- cadre shares;
- the projection horizon; and
- the selected countries.

Results that remain stable across plausible parameter combinations may be considered more robust than results that change substantially under small parameter adjustments.

18 Conclusion

The HRH Gap Tool combines historical workforce trends, demographic projections, benchmark staffing requirements and a composite health-need index to support workforce planning under alternative scenarios. The model provides a transparent and configurable framework for examining potential workforce shortages and coverage gaps.

Its outputs should be used as decision-support information rather than exact forecasts. Appropriate interpretation requires careful review of input data, parameter choices, assumptions and national policy context.

References

- Birch, S., Kephart, G., Tomblin Murphy, G., O'Brien-Pallas, L., Alder, R., & MacKenzie, A. (2009). Human resources planning and the production of health: A needs-based analytical framework. *Canadian Public Policy*, 33(Supplement 1), S1–S16. <https://doi.org/10.1097/PHH.0b013e3181b1ec0e>
- Chang, W., Cheng, J., Allaire, J. J., Sievert, C., Schloerke, B., Xie, Y., et al. (2025). *Shiny: Web application framework for R*. Posit Software. <https://shiny.posit.co/>
- GBD 2021 Diseases and Injuries Collaborators. (2024). Global burden of 288 causes of death and disease burden in 204 countries and territories, 1990–2021. *The Lancet*, 403(10440), 2100–2132. [https://doi.org/10.1016/S0140-6736\(24\)00367-2](https://doi.org/10.1016/S0140-6736(24)00367-2)
- Hall, T. L. (2001). *Human resources for health: Models for projecting workforce supply and requirements*. World Health Organization.
- Little, R. J. A., & Rubin, D. B. (2019). *Statistical analysis with missing data* (3rd ed.). John Wiley & Sons.
- Nardo, M., Saisana, M., Saltelli, A., Tarantola, S., Hoffman, A., & Giovannini, E. (2008). *Handbook on constructing composite indicators: Methodology and user guide*. OECD Publishing.
- Preston, S. H., Heuveline, P., & Guillot, M. (2001). *Demography: Measuring and modeling population processes*. Blackwell Publishers.
- R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. <https://www.r-project.org/>
- Saltelli, A., Ratto, M., Andres, T., Campolongo, F., Cariboni, J., Gatelli, D., Saisana, M., & Tarantola, S. (2008). *Global sensitivity analysis: The primer*. John Wiley & Sons.
- United Nations Department of Economic and Social Affairs, Population Division. (2024). *World population prospects 2024*. United Nations.
- van Buuren, S. (2018). *Flexible imputation of missing data* (2nd ed.). CRC Press.
- World Health Organization. (2006). *The world health report 2006: Working together for health*.
- World Health Organization. (2016). *Global strategy on human resources for health: Workforce 2030*.
- World Health Organization. (2022). *National health workforce accounts: A handbook*.
- World Health Organization. (2023a). *Health workforce support and safeguards list 2023*.
- World Health Organization. (2023b). *Workload indicators of staffing need (wisn): User's manual*.
- World Health Organization & World Bank. (2023). *Tracking universal health coverage: 2023 global monitoring report*.
- Zeileis, A., & Grothendieck, G. (2024). *Zoo: S3 infrastructure for regular and irregular time series*. CRAN. <https://CRAN.R-project.org/package=zoo>